



Charts and Graphs

DSCI 454
Week 4, Lecture 1

What are we learning this lecture?



- Charts and Graphs
 - Choosing the right chart for the right story.
 - Common mistakes and misleading visuals.



Inspired by

Storytelling with Data, Knaeflic

“Chart” vs “Graph”

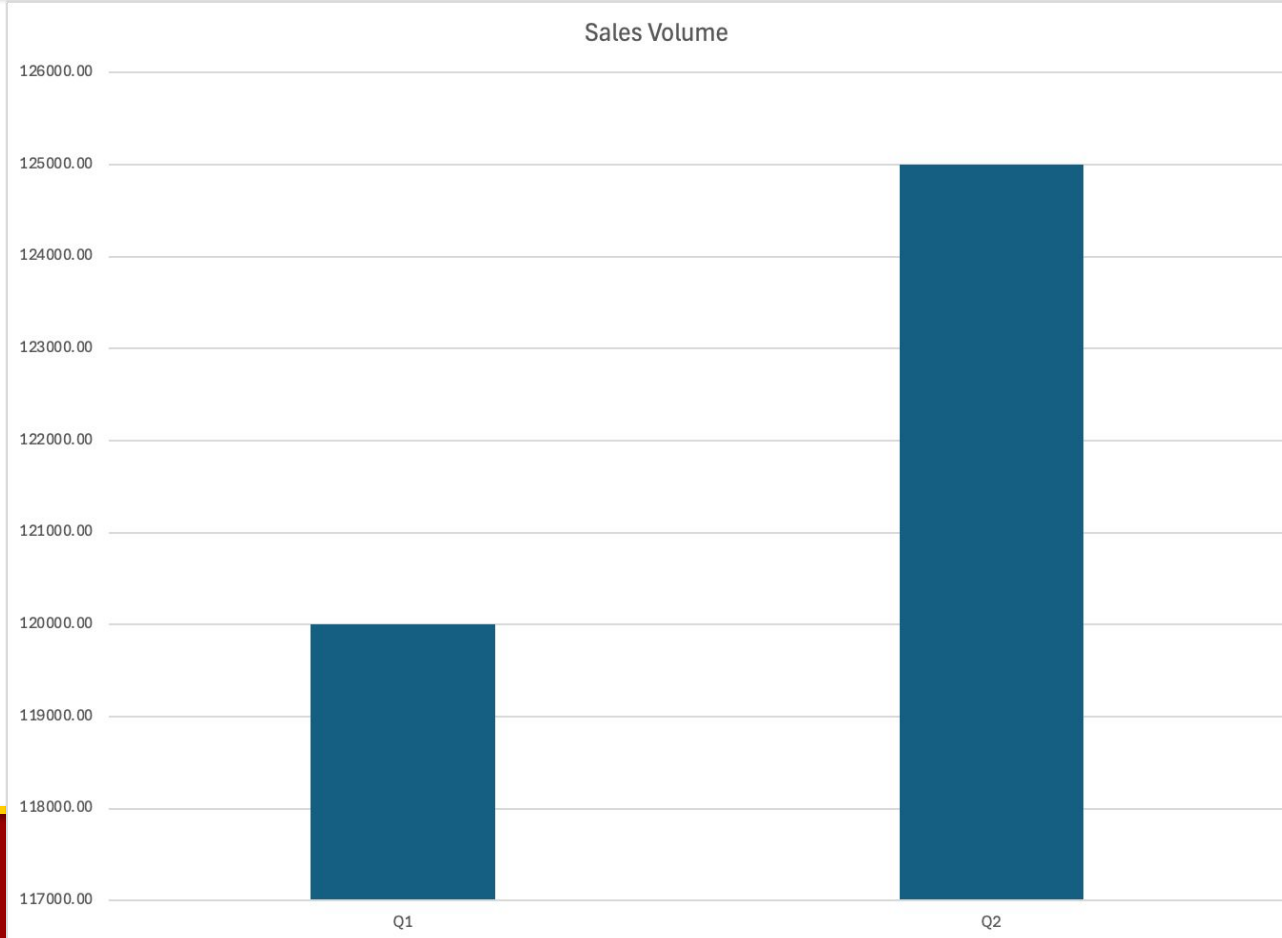


- Technically a Graph is a type of chart w/ axes e.g. y vs x w/ a hierarchy that looks like this
- Chart
 - Graph
 - Map
 - Diagram



Right Chart for Right Story

Do you even need a chart?



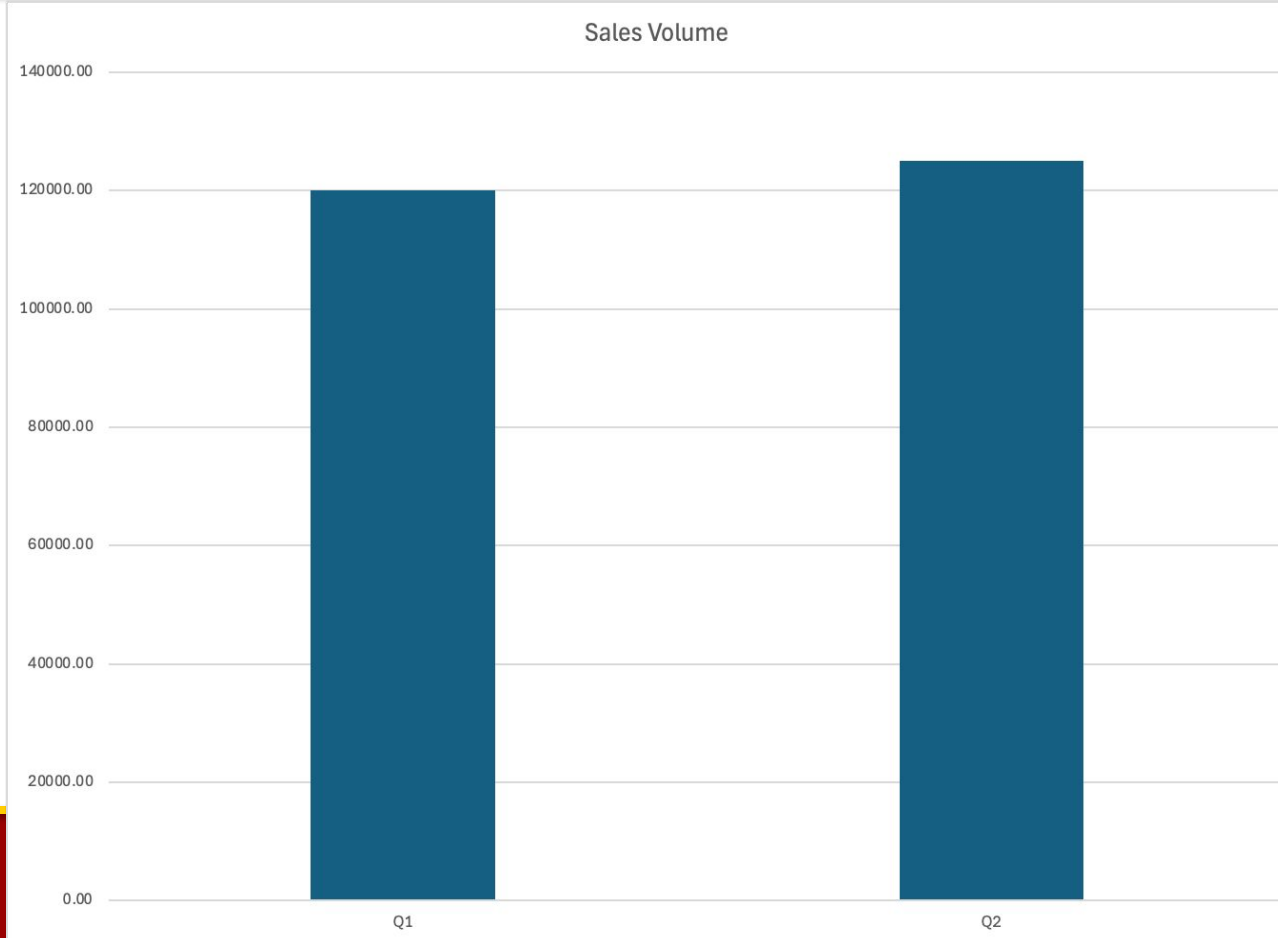
Or, in other words,

$\$125\text{K} - \$120\text{K} = \$5\text{K}$

$/ \$120\text{K} =$

“**4.2%** growth from Q1 to Q2” ...

Do you even need a chart?



Or, in other words,
“**4.2%** growth from
Q1 to Q2” ...

Reminder: Check the
y-axis min truncation
bias!

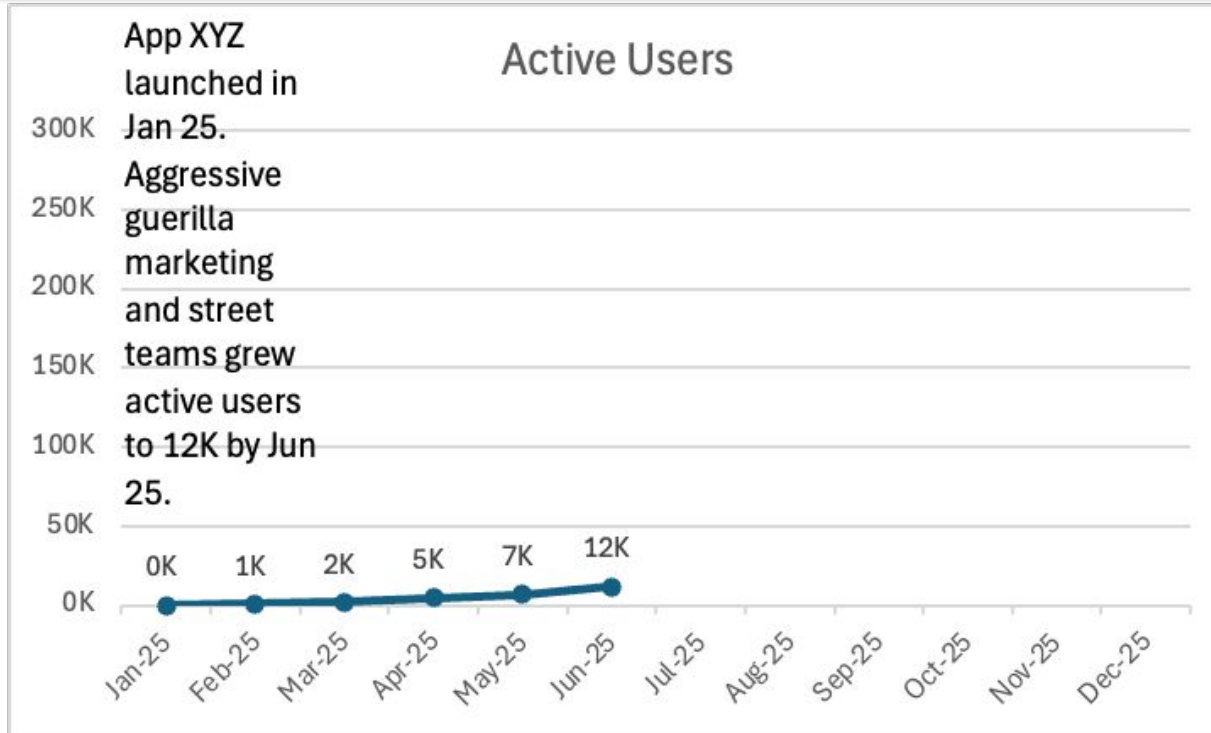
Sometimes, a
**sentence is all you
need.**

What story are you trying to tell?



- Metric / Measurement / KPI e.g. revenue, active users growth over time?
 - Line or Bar
- Geographic Expansion over time?
 - Map scatter or Choropleth
- Comparison of categories, e.g. product types, user types (demographics), options, selections, choices
 - Bar or Column
- Series of charts or Animated chart?
 - Control w/ voice track vs. Pizazz

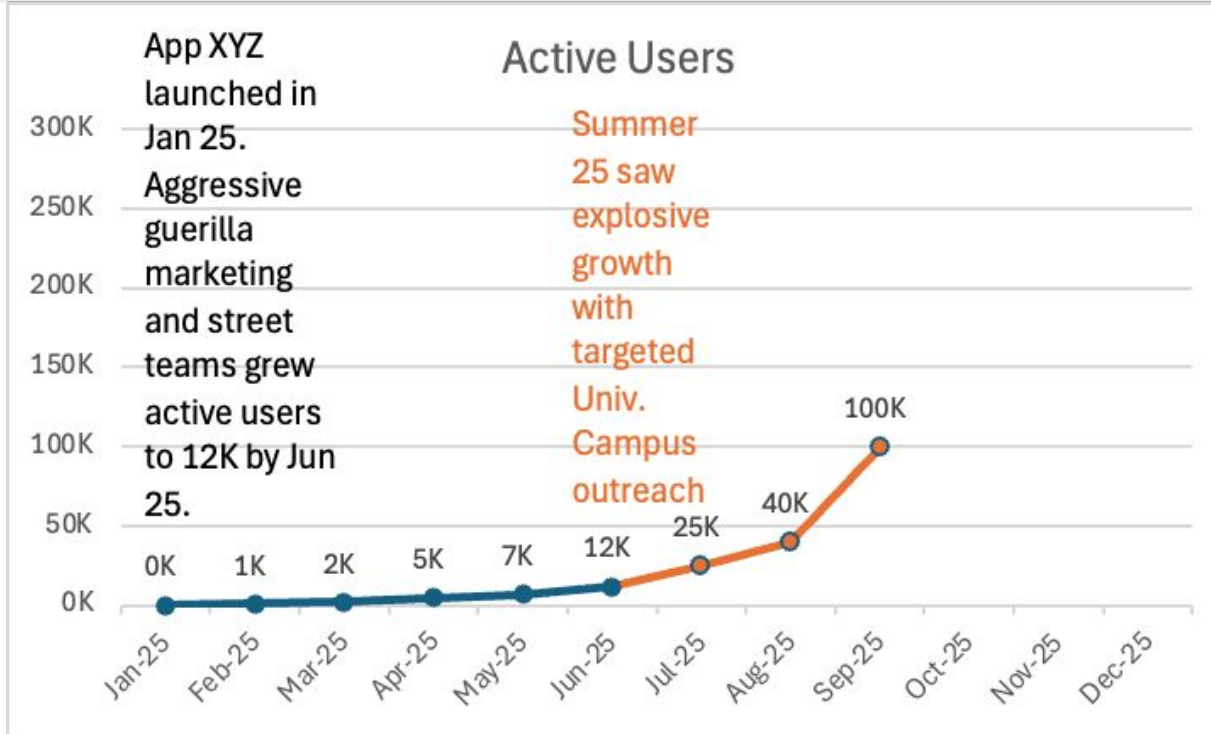
Line Growth Series



Add callouts for points or ranges to explain growth

In Excel, Format Chart to keep y max and x max values to the max of the entire data set

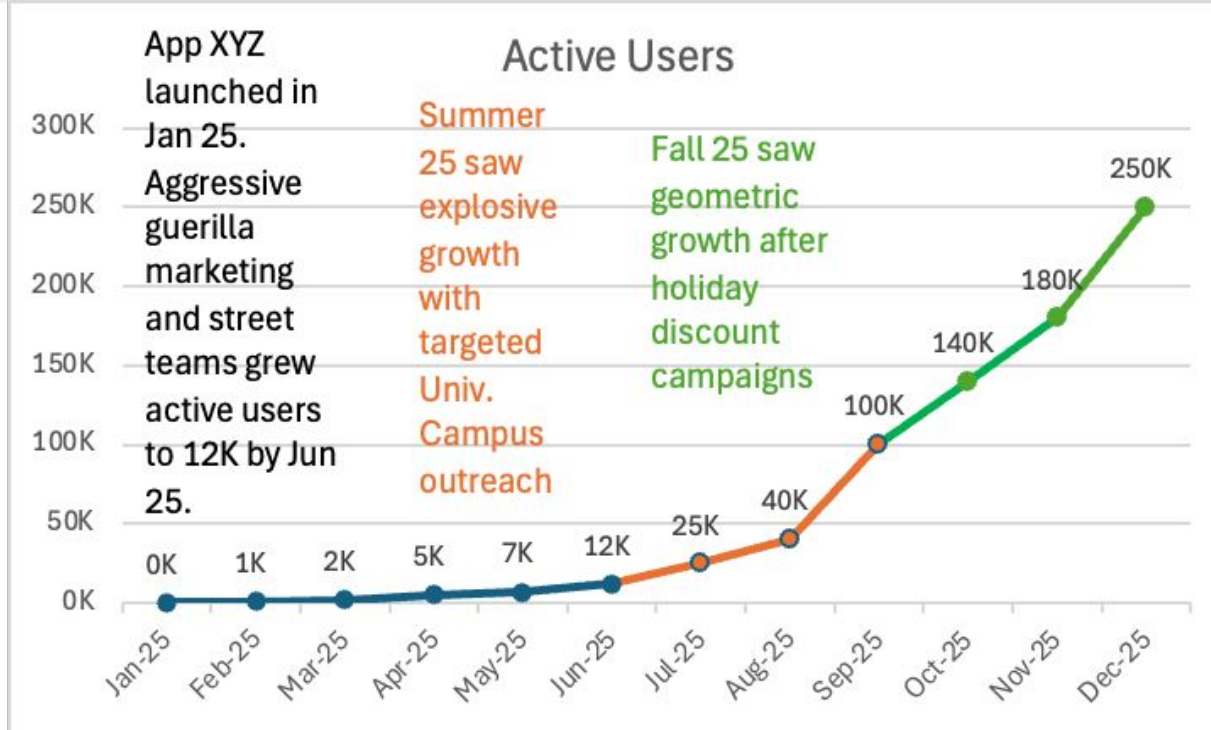
Line Growth Series



Keep the frame consistent across the series

Optionally color code points or ranges, careful not to get too busy or cluttered

Line Growth Series



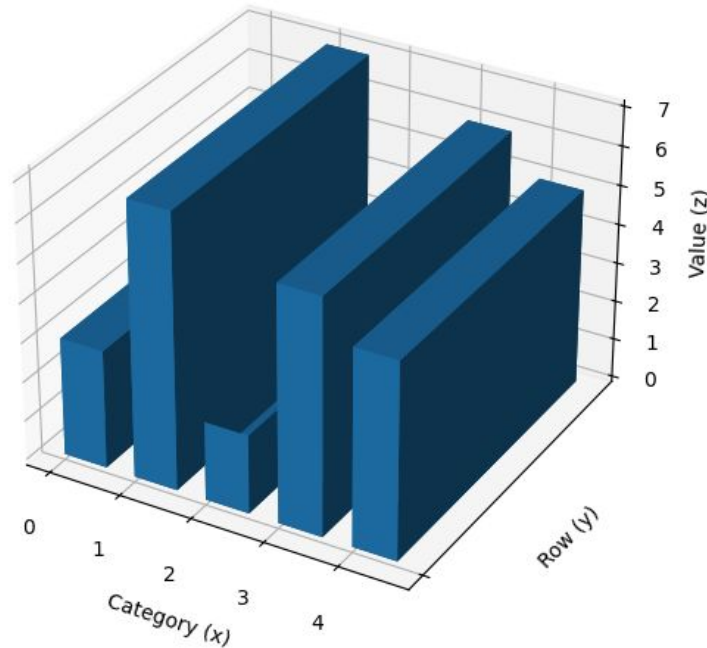
Be careful not to get too busy or cluttered (color, space, etc.)

Optionally, add a projection for Q1 2026 as a dashed line



Things to Avoid

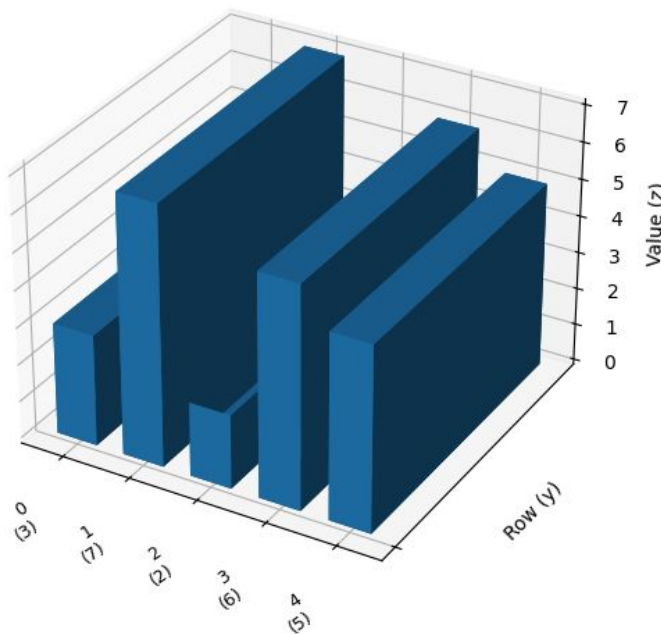
Avoid 3D



Q: What are the height values for each column?

Hard for humans to project height plane onto gridlines

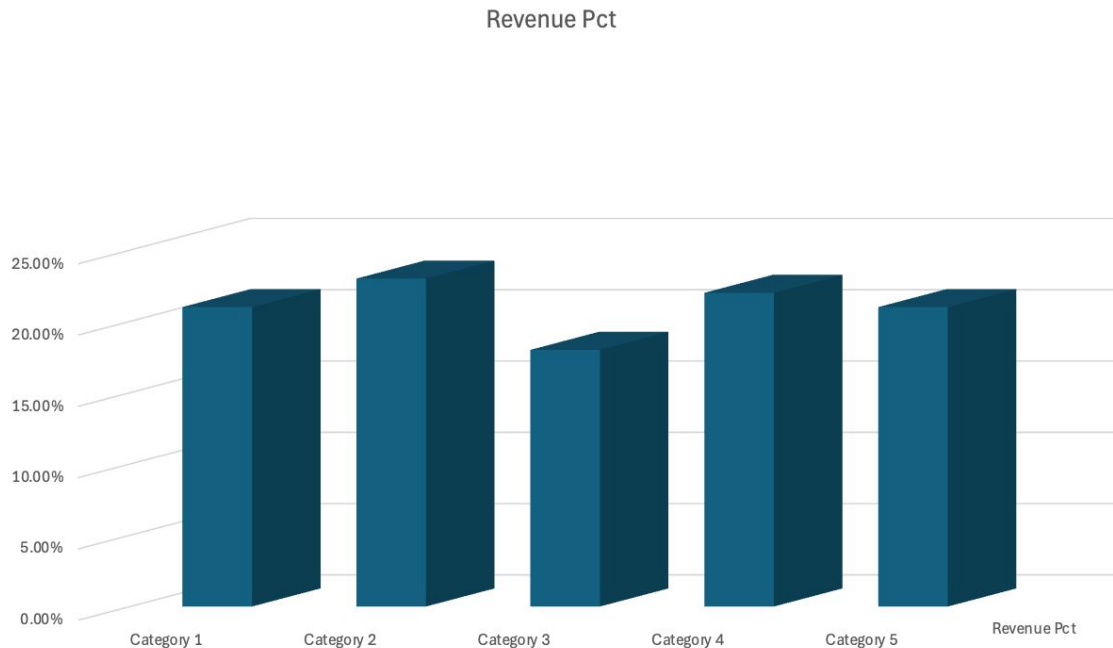
Avoid 3D



A: The heights don't necessarily line up w/ the grid lines.

Note, the grid lines for Category 0 and 2 are not even visible in this orientation.

3d Column in Excel



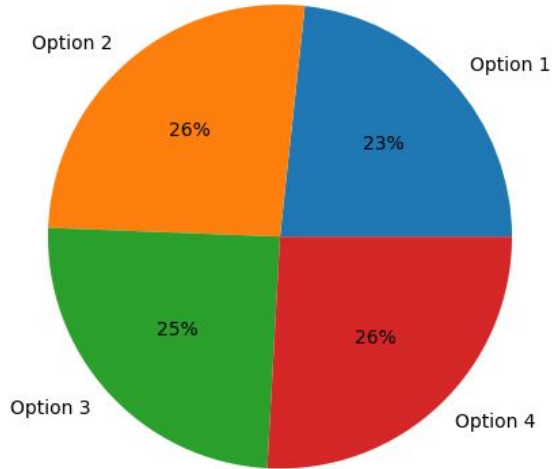
Excel and matplotlib both allow for 3d rotation

Both are hard to perceive height values from.

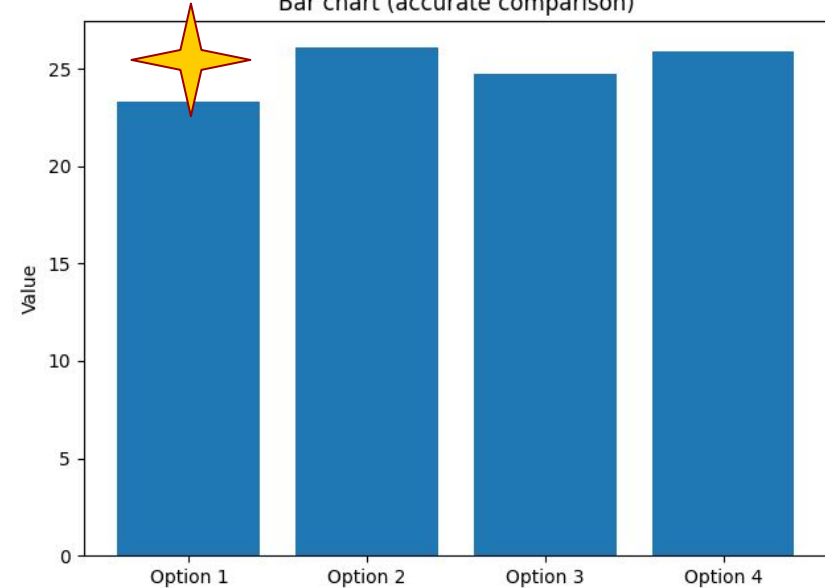
Reminder: Avoid Pie Charts



Pie chart (misleading for close values)



Bar chart (accurate comparison)

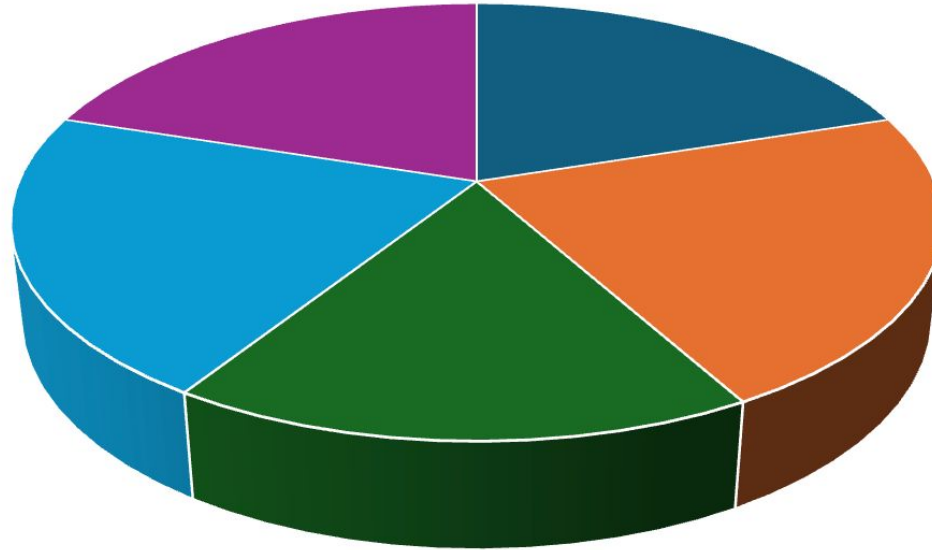


Intentional or not, a pie chart can obscure accuracy/precision of close call results clearly visible in a bar chart.

Avoid 3D Pie Charts, Double bad



Sales Data



■ Category 1 ■ Category 2 ■ Category 3 ■ Category 4 ■ Category 5

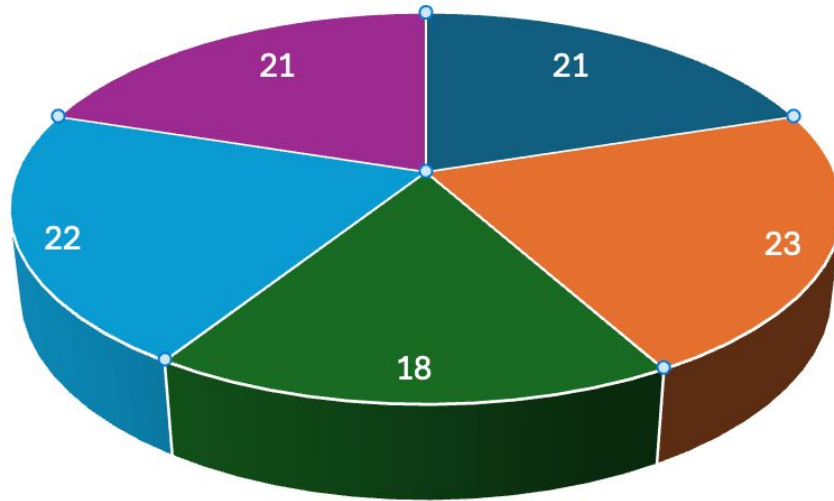
Q: Which category is biggest?

Human eyes are bad at estimating angles

Avoid 3D Pie Charts, Double bad



Sales Data



■ Category 1 ■ Category 2 ■ Category 3 ■ Category 4 ■ Category 5

A: Orange.

Did you guess correctly?

Human eyes are bad at estimating angles

Also Avoid Donuts



Revenue Pct

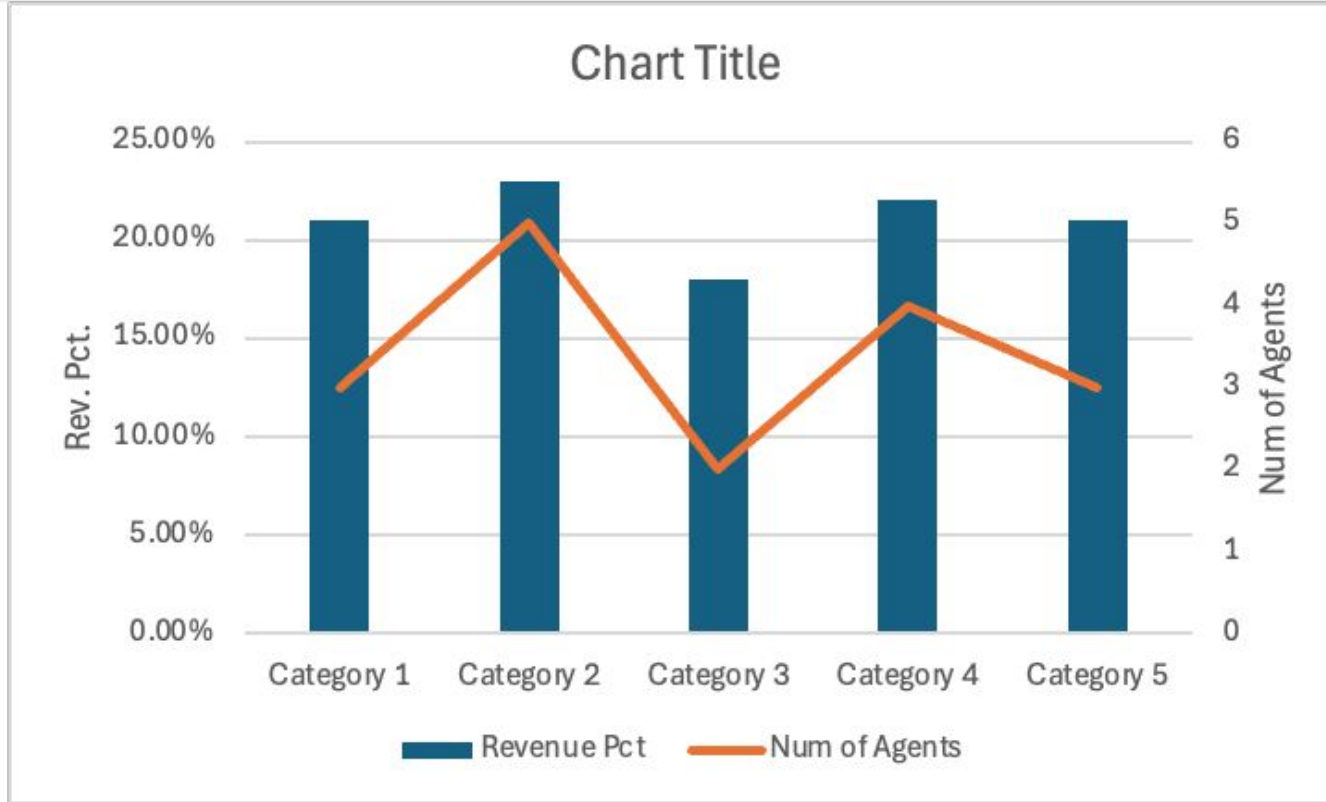


■ Category 1 ■ Category 2 ■ Category 3 ■ Category 4 ■ Category 5

For the same reason:

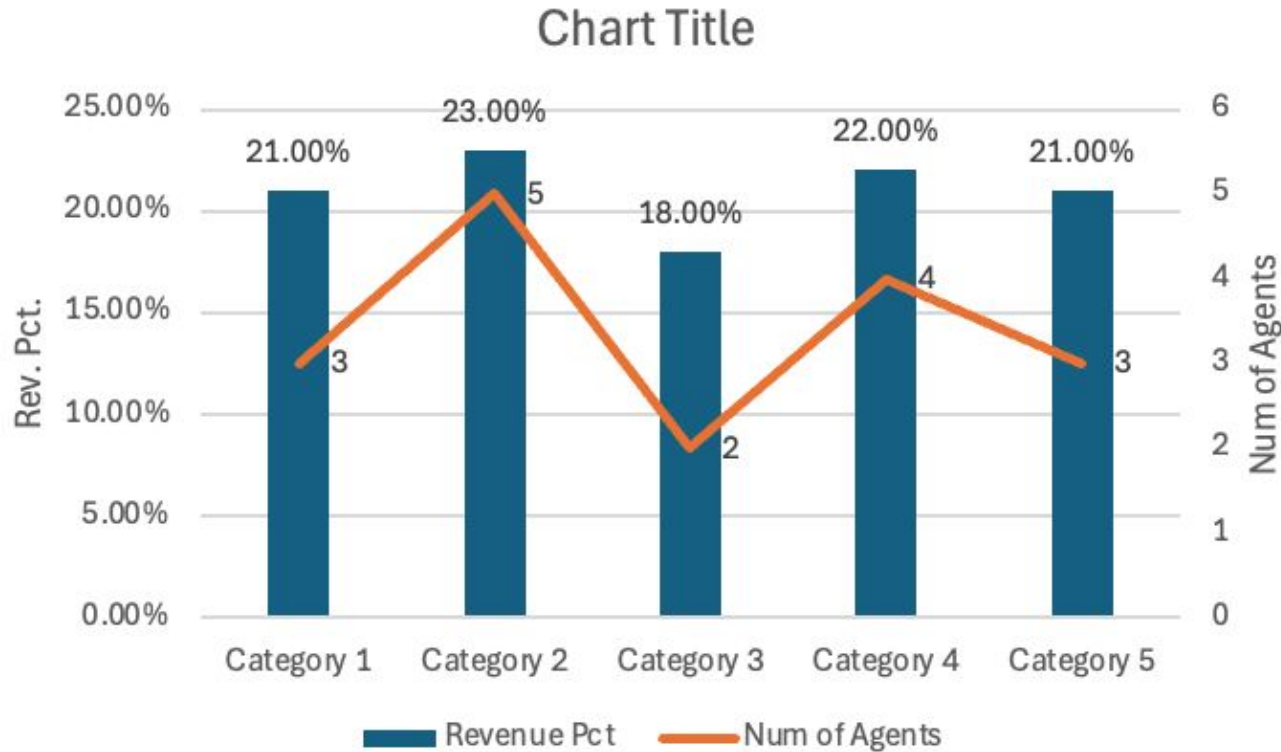
Human eyes are just as bad at estimating arc lengths as they are angles

Avoid Secondary Y-Axis (aka Combo charts)



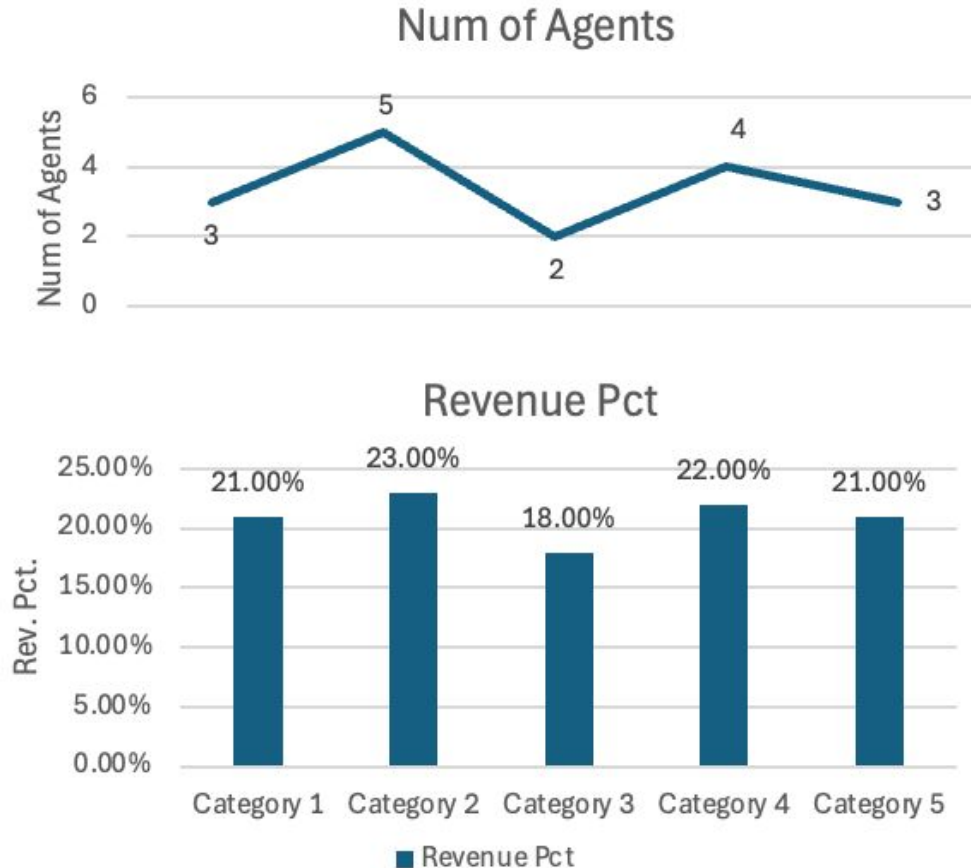
Notice how much your eyes are moving?

Secondary Y-Axis: Alt / Fix 1



Either label directly (busy / cluttered) or

Secondary Y-Axis: Alt / Fix 2



Separate Y axes and share X-Axis

Note: In Excel, this can be accomplished by making two separate plots, stacking, then removing x-axis label on top one

Questions?

